

An introduction version control

https://stirlingcodingclub.github.io/version_control

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Using version control

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- ▶ Focus on practical skills for research
 - ▶ Learn and reinforce knowledge on how to use **key skills** effectively
 - ▶ Focus on [GitHub](#) and [GitHub Desktop](#) software

Using version control

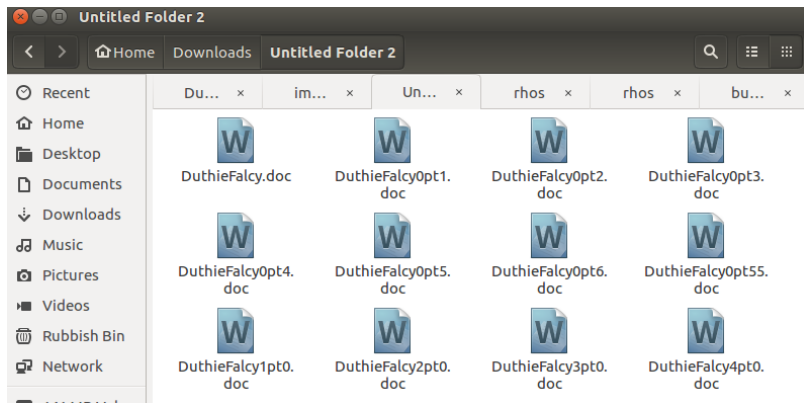
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- ▶ Focus on practical skills for research
 - ▶ Learn and reinforce knowledge on how to use **key skills** effectively
 - ▶ Focus on [GitHub](#) and [GitHub Desktop](#) software
- ▶ Hands-on practice setting up and using version control in your own work with [accompanying notes for guidance](#)

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Rough outline of version control

1. What is version control, and why use it?
2. Getting set up – good file management
3. Setting up [GitHub](#)
4. The [GitHub Desktop](#) tutorial
5. Independent work using version control

What is version control, and why use it?



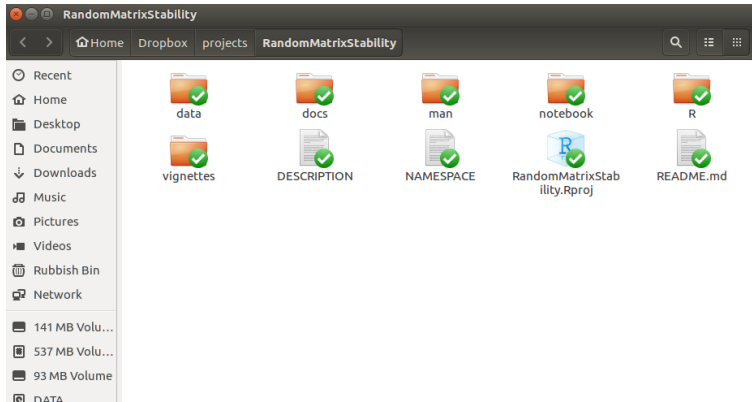
What version control software does

- ▶ Software that records changes you make to files over time
 - ▶ Manage different *versions* of files (no need to 'Save As...')
 - ▶ Recover old files, keep track of file changes
 - ▶ Collaborate with others on shared files

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- ▶ **Put more intuitively**, version control takes a snapshot in time (called a '**commit**') of all the files in one of your folders (called '**repositories**')
 - ▶ Visualise changes to your files over time
 - ▶ Look at the differences between file versions
 - ▶ Record who changed files, and what they changed

Inside of a project on version control



Folders (a.k.a, 'repositories') include all data files, R code, notes, manuscript drafts, etc.

Full annotated timeline of folder changes (GitHub)

Commits on Mar 1, 2019

Some work on the SI

 **bradduthle** committed on 1 Mar 2019 ✓



5d29331



Major restructure and revision of the Discussion to compare to Gibbs

 **bradduthle** committed on 1 Mar 2019 ✓

..et al.



a40ede5



Commits on Feb 27, 2019

Edit the manuscript up to Reviewer 2 specific comment 3 -- these are

 **bradduthle** committed on 27 Feb 2019 ✓

..next on the list, specifically the new Discussion paragraph



d411fd5



Commits on Feb 22, 2019

First pass, fix some notation

 **bradduthle** committed on 22 Feb 2019 ✓



39f854b



Add some to the abstract to emphasise finite systems

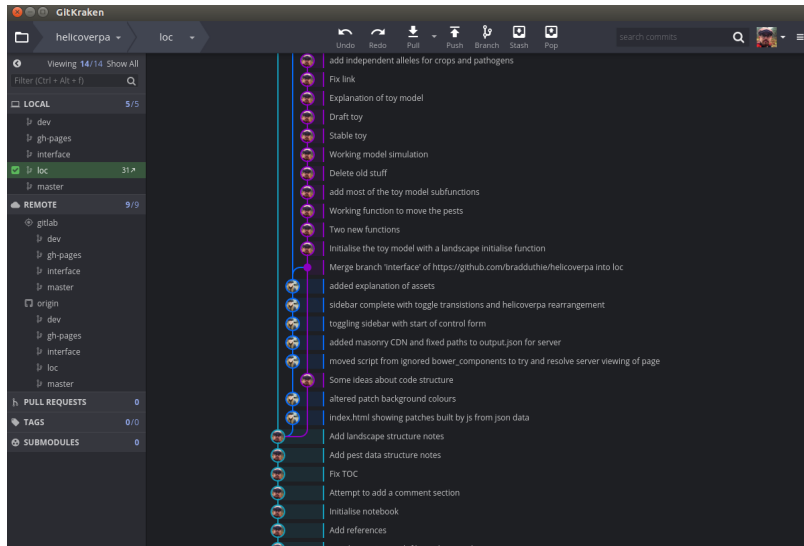
 **bradduthle** committed on 22 Feb 2019 ✓



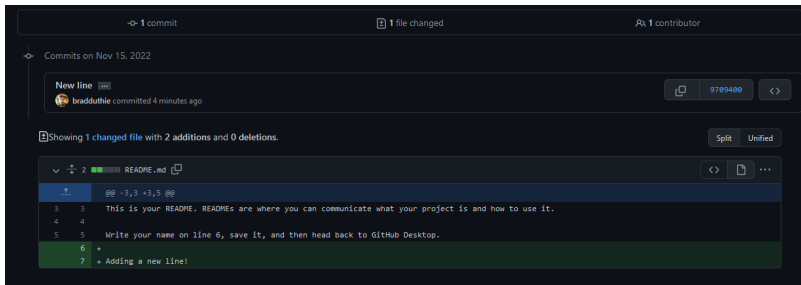
612aa4b



Parallel versions ('branches') of a folder



Clear breakdown of what has changed



The screenshot shows a GitHub commit diff interface. At the top, it indicates "1 commit", "1 file changed", and "1 contributor". Below this, it shows the commit message "New line" and the committer "bradduthie" who committed "4 minutes ago". The diff summary states "Showing 1 changed file with 2 additions and 0 deletions." The file being changed is "README.md". The diff shows three lines of code with line numbers 3, 4, and 5 on the left, and line numbers 6 and 7 on the right. Line 3 is a separator line with three backticks and spaces. Line 4 is a comment. Line 5 is a comment. Line 6 is a new line added, indicated by a green background and a "+" sign. Line 7 is another new line added, also indicated by a green background and a "+" sign.

1 commit 1 file changed 1 contributor

Commits on Nov 15, 2022

New line
bradduthie committed 4 minutes ago

Showing 1 changed file with 2 additions and 0 deletions. Split Unified

2 README.md

```
@@ -3,3 +3,5 @@  
3 3 This is your README. READMEs are where you can communicate what your project is and how to use it.  
4 4  
5 5 Write your name on line 6, save it, and then head back to GitHub Desktop.  
6 +  
7 + Adding a new line!
```

Version control makes science easier

- ▶ **Organises files** by avoiding 'save as' multiple versions
 - ▶ analysis_1.R
 - ▶ analysis_2.R
 - ▶ analysis_FINAL.R
 - ▶ analysis_FINAL_no_really_this_time.R

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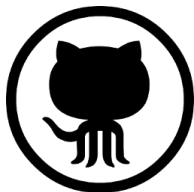
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- ▶ **Gives peace of mind** to experiment by removing any fear of breaking something that you know works

Version control can help open science



- ▶ Transparent record of data collection, analysis, and writing
- ▶ Record publicly available on [GitHub](#), [Bitbucket](#), or [GitLab](#)
- ▶ GitHub repository can be copied, reproduced, and discussed
- ▶ [git](#) and GitHub can track individual contributions to a project

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- ▶ Works across platforms
 - ▶ Windows
 - ▶ Linux
 - ▶ Mac
- ▶ Invented by [Linus Torvalds](#)